TIDES: Time-Parallel Dual-Sparse Packing for Efficient Systolic SNN Acceleration

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V. Evaluation

In this section, we first detail our evaluation methodology. Next, we justify the weight-stationary dataflow over input- and output-stationary alternatives. We then evaluate the performance, energy and area of TIDES when applied to SNN models. Additionally, we report the compression achieved by packing and illustrate the reduction in memory traffic.

A. Methodology

Evaluation Models and Datasets. We evaluate TIDES on spiking CNNs (VGG16 and ResNet18) on CIFAR-10 with $T = 8$ timesteps. We obtain the models from the open-source NeuroFlex repository [1], which converts INT8-quantized ANNs into PASCAL SNNs whose spiking execution reproduces the quantized ANN exactly, and evaluate them using PyTorch; the dense VGG16 reaches 95.62% test accuracy. To expose weight sparsity, we prune the weights to 90% sparsity with magnitude pruning and fine-tune the models, reaching – and – accuracy on VGG16 and ResNet18, respectively. Then, we obtain the input spike trains and INT8 weights of every convolutional and fully connected layer and utilize this information in our experiments. The first layer, which receives the analog image, is excluded for all designs, and a layer that follows average pooling is executed exactly on the four pre-pooling spikes of each pooled input. For sensitivity studies, we additionally generate synthetic layers of the same shapes with independently set weight sparsity, fraction of silent neurons and per-timestep firing probability.

TIDES Implementation. We developed a cycle-level simulator that takes the spike trains and weights of the model as input, modeling the behavior of the packer, the PE array, the edge unit and the LIF units to estimate the runtime performance of TIDES. We implement all components in Verilog; in a closed-loop Verilator [2] testbench, the RTL reproduces the partial sums, output spikes and cycle count of the simulator exactly on every layer of both networks at batch sizes 1 and 16. The components are synthesized with Yosys and ABC [3] on the Nangate 45 nm open cell library [4], and OpenSTA [5] provides timing and static/dynamic power from gate-level switching activity of real layers. We use CACTI 7.0 [6] to evaluate the area and power of the on-chip buffers in a 45 nm process, with access counts from the simulator. We use the DRAM energy per bit derived with DRAMsim3 [7] in [8] to estimate the off-chip DRAM energy. Tbl. I shows the setup of the TIDES architecture.

TABLE I
TIDES Architecture Setup.

Parameter	Value
Tile Size	$R = 4$ inputs; output 64 filters $\times$ 128 pixels
Packing	$F = 4$ filters/column; $G = 4$ positions/group
Computation Array	4×4 PEs $\times$ 8 adders; 16 LIF units
Edge Unit	4 FIFOs $\times$ 80 entries; 4 RMW/column/cycle
On-chip Buffer Size	192 KB Output; 64 KB Staging; 4 KB Group
DDR4 DRAM	4 Channels, 64 GB/s

Baselines. We compare with the SNN accelerators PTB [9], SpikeX [10], SATO [11], SpikeFlow [12], LoAS [13] and Prosperity [8], which we discussed in Sec. II. We compare their configuration and performance on VGG16 with CIFAR-10 in Tbl. II. For a fair comparison, we keep the frequency and technology the same for all baselines and TIDES: the core of each baseline is implemented in RTL and taken through the same synthesis and power flow, and its on-chip buffers, sized as in its paper, are evaluated with CACTI 7.0 in the same process. We define operations (OP) consistently across all baselines and TIDES, where each OP corresponds to an accumulation for a '1' element in the spike bit matrix. We simulate PTB, SATO and Prosperity with the official simulators released with Prosperity [8], and SpikeX, SpikeFlow and LoAS in our framework. On our VGG16 traces, the official simulators give Prosperity a 9.6$\times$ speedup over PTB, close to the 9.4$\times$ reported in [8].

References

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TABLE II
COMPARISON OF TIDES WITH BASELINES ON VGG16 (CIFAR-10). AREA INCLUDES THE ON-CHIP BUFFERS.

	Freq. (MHz)	Tech. (nm)	Area (mm^2)	Throughput (GOP/s)	Energy Efficiency (GOP/J)	Area Efficiency (GOP/s/mm^2)
PTB [9]	500	45	1.232	– (1.00×)	– (1.00×)	– (1.00×)
SpikeX [10]	500	45	0.747	–	–	–
SATO [11]	500	45	0.354	–	–	–
SpikeFlow [12]	500	45	0.207	–	–	–
LoAS [13]	500	45	1.478	–	–	–
Prosperity [8]	500	45	1.035	–	–	–
TIDES	500	45	2.128	–	–	–

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